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In high dimensions, even this simple shape reveals **surprising and counter-intuitive geometric behavior.**

High-Dimensional Cube (examples)

- let's first take the case of the unit cube $\left[-\frac{1}{2}, \frac{1}{2}\right]^d$ centered at the origin in dimension $d = 2$ (i.e., a square)
- the distance from the origin $(0,0)$ to a corner $\left(\frac{1}{2}, \frac{1}{2}\right)$ is $\sqrt{\frac{1}{2^2} + \frac{1}{2^2}} = \frac{1}{\sqrt{2}} < 1$, which means that the square lies inside the unit sphere

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- we can repeat the calculation for $d = 4$, where we find that the corners of the hypercube intersect with the sphere since:

$$\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1$$

High-Dimensional Cube (distance to corner)

- in arbitrary dimension d , the distance between the origin and a corner of the hypercube is equal to $\sqrt{\sum_{i=1}^d \frac{1}{2^2}} = \frac{\sqrt{d}}{2}$

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- in arbitrary dimension d , the distance between the origin and a corner of the hypercube is equal to $\sqrt{\sum_{i=1}^d \frac{1}{2^2}} = \frac{\sqrt{d}}{2}$
- as shown in the figure, this implies that the corners of the hypercube lie outside the hypersphere for $d > 4$

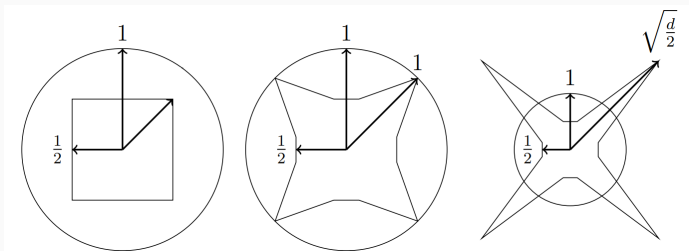


Figure 1: Relation between hypersphere and hypercube in dimensions 2,4, and arbitrary $d > 4$.

The Volume of the Cube is Located in Its Corners (statement)

- we formally prove the behavior shown in the previous slide's figure where we observed that the corners lie outside the hypersphere
- we will consider the cube $[-1, 1]^d$ but the same argument can be made with $[-\frac{1}{2}, \frac{1}{2}]^d$ (as shown in the previous slide's figure)

Proposition 1

most of the volume of the high-dimensional cube is located in its corners

The Volume of the Cube is Located in its Corners (sketch of the proof)

proof:

- consider a point $\mathbf{x} \in [-1, 1]^d$ chosen uniformly from the hypercube (i.e. each coordinate x_i is drawn uniformly at random from the interval $[-1, 1]$)
- in order to establish whether this points is more likely to be at a corner, we will bound the probability that this random point is **inside the hypersphere**. A point $\mathbf{x}$ lies **inside the hypersphere** if $\|\mathbf{x}\|_2 \leq 1$.
- Since for $d > 4$ the corners of the hypercube lie outside the hypersphere estimating, $\Pr(\|\mathbf{x}\|_2 \leq 1)$ will tell us how much volume is near the corners.

The Volume of the Cube is Located in its Corners (moments)

To understand the distribution of $\|\mathbf{x}\|_2^2$, we start by computing its mean and variance:

$$\mathbb{E}[x_i^2] = \frac{1}{2} \int_{-1}^1 z^2 dz = \frac{1}{3} \quad \Rightarrow \quad \mathbb{E}[\|\mathbf{x}\|_2^2] = \frac{d}{3}.$$

and

$$\text{Var}(x_i^2) = \frac{1}{2} \int_{-1}^1 z^4 dz - \frac{1}{9} = \frac{4}{45} \leq \frac{1}{10} \quad \Rightarrow \quad \text{Var}[\|\mathbf{x}\|_2^2] \leq \frac{d}{10}.$$

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However, moments alone don't show how tightly $\|\mathbf{x}\|_2^2$ concentrates around its mean. To quantify this, we use a **concentration inequality**.

From variance to concentration: why we need Chernoff

We want a bound on the probability of large deviations:

$$\Pr \left(\left| \|\mathbf{x}\|_2^2 - \mathbb{E} \|\mathbf{x}\|_2^2 \right| > t \right).$$

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To get such a bound, we use a general-purpose tool for sums of independent bounded random variables: **Chernoff's inequality**.

Chernoff's inequality

Theorem 1 (Chernoff's Inequality)

Let $Z_1, \dots, Z_d$ be independent random variables with zero mean such that a.s. for all i $|Z_i| \leq 1$ a.s.. Also define $\sigma_i := \mathbb{E}[Z_i^2]$ and $\sigma^2 := \sum_{i=1}^d \sigma_i^2$. Then, for any $t > 0$

$$\Pr\left(\sum_{i=1}^d Z_i \leq -t\right) \leq \max\left(e^{-\frac{t^2}{4\sigma^2}}, e^{-\frac{t}{2}}\right)$$

and

$$\Pr\left(\sum_{i=1}^d Z_i \geq t\right) \leq \max\left(e^{-\frac{t^2}{4\sigma^2}}, e^{-\frac{t}{2}}\right)$$

Applying Chernoff's inequality

Recall: A point $\mathbf{x}$ lies inside the unit sphere if $\|\mathbf{x}\|_2^2 = \sum_{i=1}^d x_i^2 \leq 1$.

We apply Chernoff's inequality to

$$Z_i := x_i^2 - \mathbb{E}[x_i^2]$$

Notice that

$$\sum_{i=1}^d Z_i = \sum_{i=1}^d (x_i^2 - \mathbb{E}[x_i^2]) = \|\mathbf{x}\|_2^2 - \mathbb{E}\|\mathbf{x}\|_2^2 = \|\mathbf{x}\|_2^2 - \frac{d}{3}.$$

Thus

$$\begin{aligned} \Pr(\|\mathbf{x}\|_2^2 \leq 1) &= \Pr(\|\mathbf{x}\|_2^2 - \mathbb{E}\|\mathbf{x}\|_2^2 \leq 1 - \mathbb{E}\|\mathbf{x}\|_2^2) \\ &= \Pr\left(\|\mathbf{x}\|_2^2 - \mathbb{E}\|\mathbf{x}\|_2^2 \leq 1 - \frac{d}{3}\right) \\ &\leq \exp\left(-\frac{\left(\frac{d}{3} - 1\right)^2}{\frac{4d}{10}}\right) \leq \exp\left(-\frac{d}{10}\right) \end{aligned}$$

Conclusion: Volume Concentrates in the Corners

As $d \rightarrow \infty$,

$$\Pr(\|\mathbf{x}\|_2 \leq 1) \leq e^{-d/10} \rightarrow 0.$$

Hence, a uniformly random point in $[-1, 1]^d$ almost surely lies *outside* the inscribed sphere.

Therefore, the volume of the hypercube is concentrated near its corners. $\square$

What happens when we sample from a high-dimensional Gaussian?

Gaussians in High Dimension

What happens when we sample from a high-dimensional Gaussian?

A familiar object in data science

Spherical Gaussians $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ appear everywhere:

- feature vectors in high-dimensional models,
- noise models in inverse problems,
- latent variables in generative networks.

But how does their geometry change with d ?

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- one might expect $\|\mathbf{x}\|$ to vary widely – but in fact it is almost deterministic for large d
- we will see that the distribution of $\|\mathbf{x}\|$ **concentrates sharply around** $\sqrt{d}$:

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- intuitively, this means that we can **approximate** spherical Gaussians $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ by a uniform distribution on a thin spherical shell of radius $\sqrt{d} \pm \mathcal{O}(1)$ (why?)

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- Hence, by independence,

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- The probability decays **exponentially fast** in the dimension d .

Conclusion: $\|\mathbf{x}\|_2^2$ is overwhelmingly close to its mean d . **Most Gaussian samples lie on a thin shell.**

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Theorem 2 (Gaussian Annulus Theorem)

Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. Then, for any $c > 0$,

$$\Pr \left(\sqrt{d-1} - c \leq \|\mathbf{x}\|_2 \leq \sqrt{d-1} + c \right) \geq 1 - \frac{4}{c^2} e^{-c^2/4}.$$

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As d grows, the Gaussian mass collapses into an ultra-thin spherical shell.

- We now apply the Gaussian Annulus Theorem to a problem in data science.
- **Can we separate samples coming from two different Gaussians?**

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- Consider two d -dimensional spherical Gaussians:

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whose means are separated by a distance

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- We observe only samples from the mixture and want to infer:

From which Gaussian did a given sample come?

(Source: [CMU lecture notes](#).)

Typical Distances Between Gaussian Samples

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- For $\mathbf{x}, \mathbf{y}$ drawn independently from the same Gaussian, each lies close to the surface of a sphere of radius $\sqrt{d} \pm \mathcal{O}(1)$. Hence:

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- For $\mathbf{w} \sim G_1$ and $\mathbf{z} \sim G_2$, the centers are separated by δ , giving:

$$\|\mathbf{w} - \mathbf{z}\| \approx \sqrt{\delta^2 + 2d} \pm \mathcal{O}(\sqrt{d}).$$

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- Expanding both sides shows that the inequality holds when

$$\delta \gtrsim d^{1/4}.$$

- Thus, as the dimension grows, the separation between the means must grow faster than $d^{1/4}$ to keep the clusters distinguishable.

Example: Separating Gaussians (separability condition)

- therefore, we can separate points from different Gaussians if

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which is equivalent to

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- and this holds if δ grows at least as fast as $d^{1/4}$

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random projections, distances, and averages all become nearly deterministic.

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- **Concentration of measure is the unifying principle:**
random projections, distances, and averages all become nearly deterministic.
- **Why it matters for data science:**
these effects explain the stability of random embeddings, the behavior of high-dimensional Gaussians, and why geometry can dominate intuition.